LLM Prediction Of Treatment Effects: Elicitation, Augmentation, And Model Dependence

Validation and learning-wave results, benchmarked against elastic net and a control-equals-treatment null.

Baseline GPT-4.1 is stronger than many augmentation variants on validation.

Reasoning helps more reliably than most augmentation tweaks.

Augmentation changes implied CONFIG-level beliefs, not just error metrics.

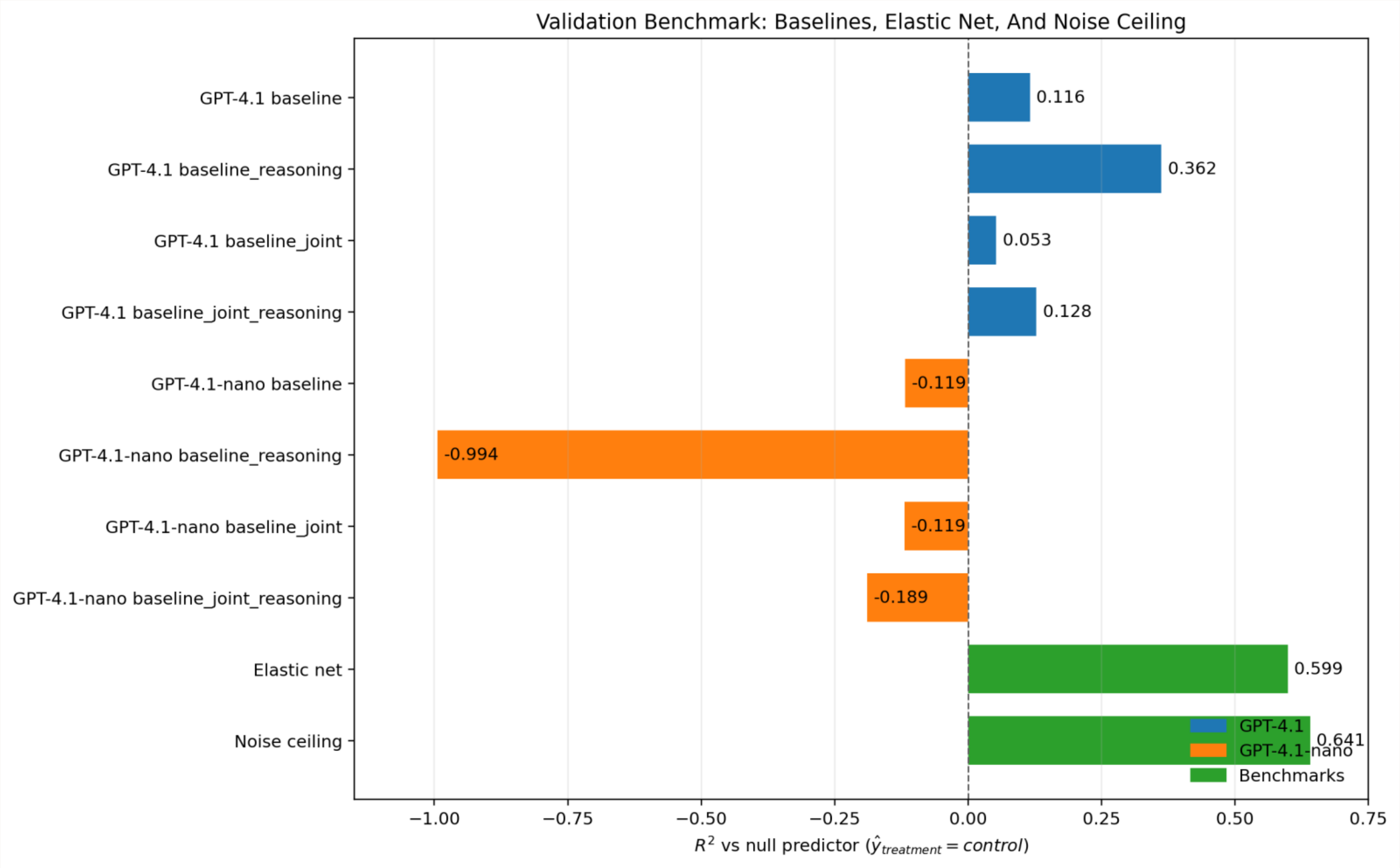
GPT-4.1-nano does not preserve the main pattern cleanly.

Baseline:

- GPT-4.1
- no augmentation (only information about the 14 design parameters + efficiency value in the control game)
- eliciting one prediction at a time
- directly outputting an integer prediction without “reasoning” process

Benchmark Before Augmentation

Validation R^2 uses the control-equals-treatment null as the denominator baseline.



GPT-4.1 baseline R^2 = 0.116; GPT-4.1 baseline_reasoning improves to 0.362.

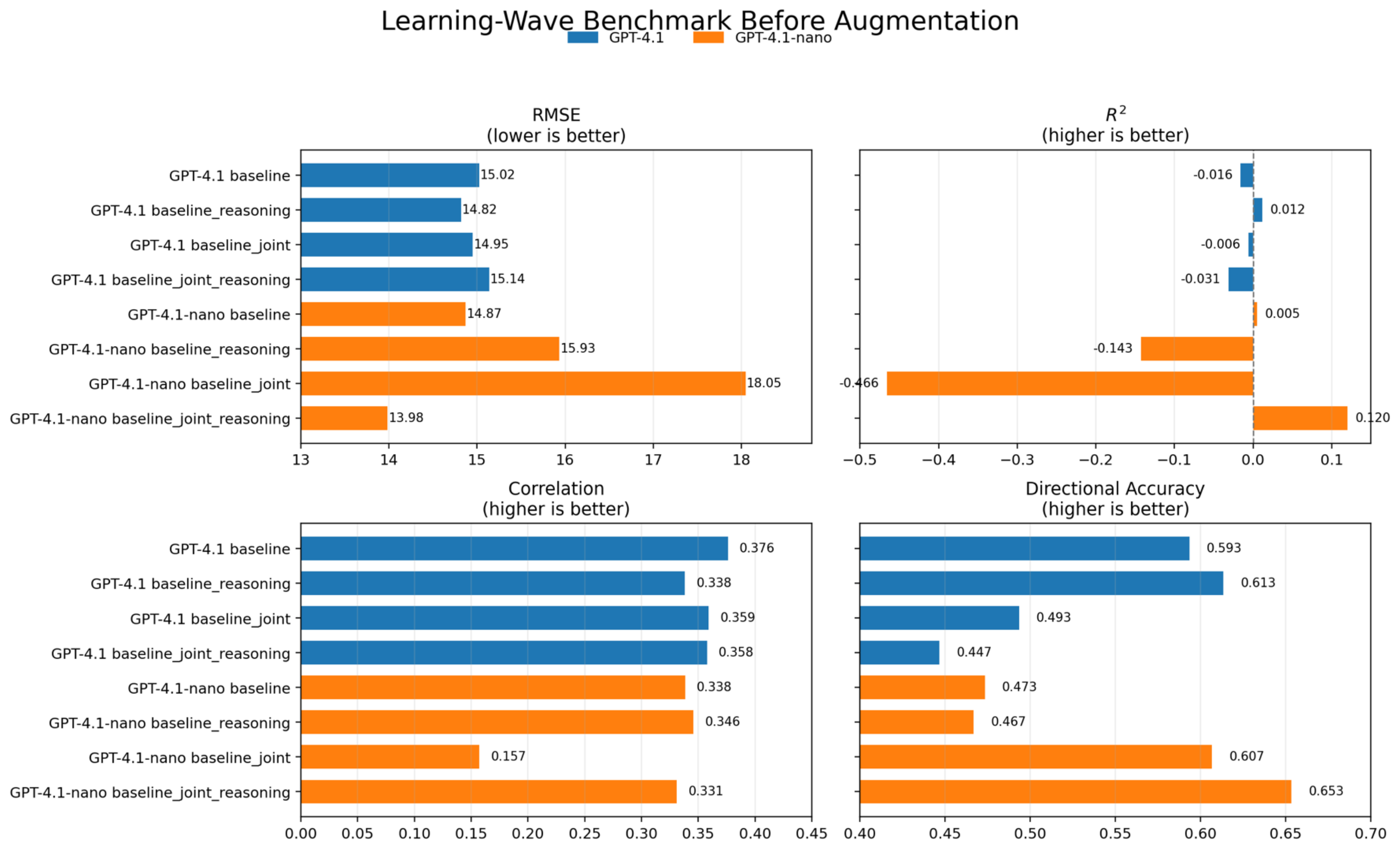
Elastic net reaches R^2 = 0.599; the noise ceiling is 0.641.

All GPT-4.1-nano baselines are at or below zero on validation.

Elastic net is much closer to the ceiling than any non-augmented LLM baseline.

Benchmark Before Augmentation (Learning Wave)

Baseline-only comparison on the learning wave



R^2 uses the control-equals-treatment null on the learning wave.

GPT-4.1 is fairly flat across elicitation modes on learning; baseline_reasoning is best on RMSE, baseline_reasoning and baseline are best on directional accuracy/correlation tradeoffs.

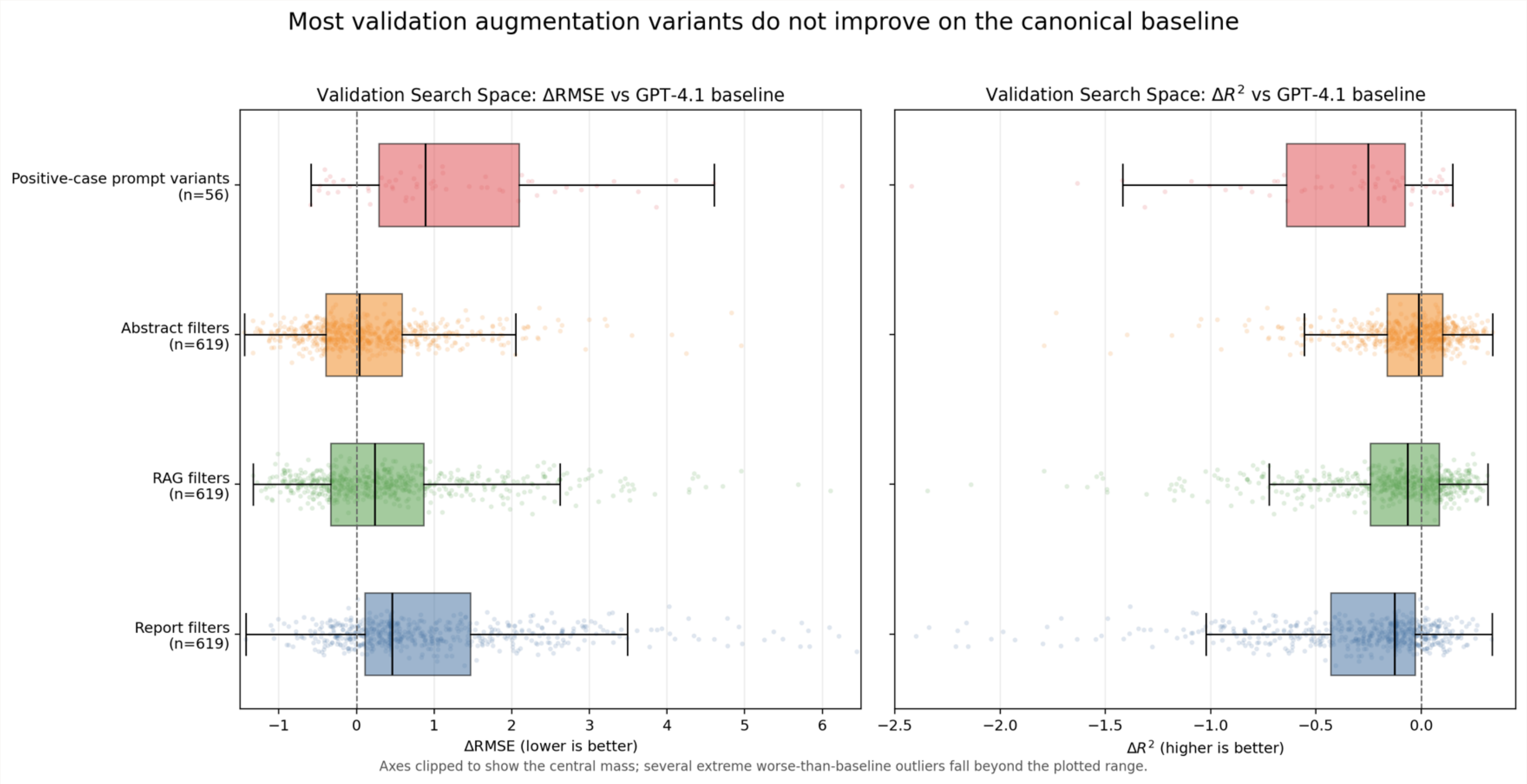
GPT-4.1-nano behaves differently: joint_reasoning is the strongest nano baseline on RMSE and directional accuracy.

Unlike validation, the learning-wave ordering is not dominated by a single clear baseline variant.

On the learning wave, the baseline ordering changes and GPT-4.1-nano no longer behaves like a weaker copy of GPT-4.1.

Main Validation Result

This summary includes the larger report/RAG/abstract search space plus the newer positive-case prompt variants.



Report filters: median ΔRMSE +0.46, only 20% of variants beat baseline on point estimate.

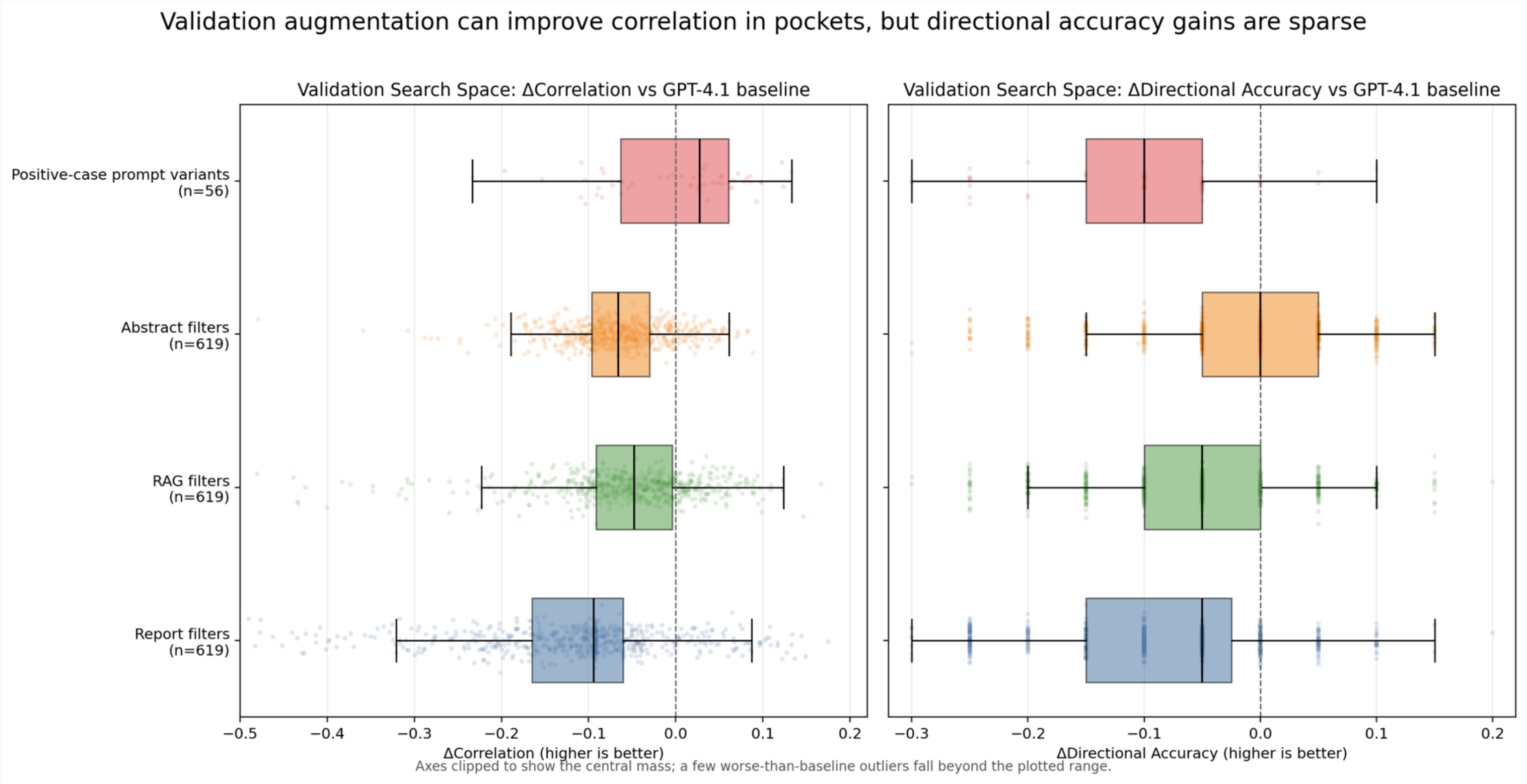
RAG filters: median ΔRMSE +0.24; Abstract filters: median +0.04.

Positive-case prompt variants are worse on average: median ΔRMSE +0.89 and median ΔR^2 -0.25.

Typical validation augmentation variants do not improve on the canonical GPT-4.1 baseline.

Validation: Correlation And Directional Accuracy

A few augmentation pockets improve correlation, but directional-accuracy gains are rare.



Report, RAG, and abstract filter families all have negative median Δcorrelation on validation.

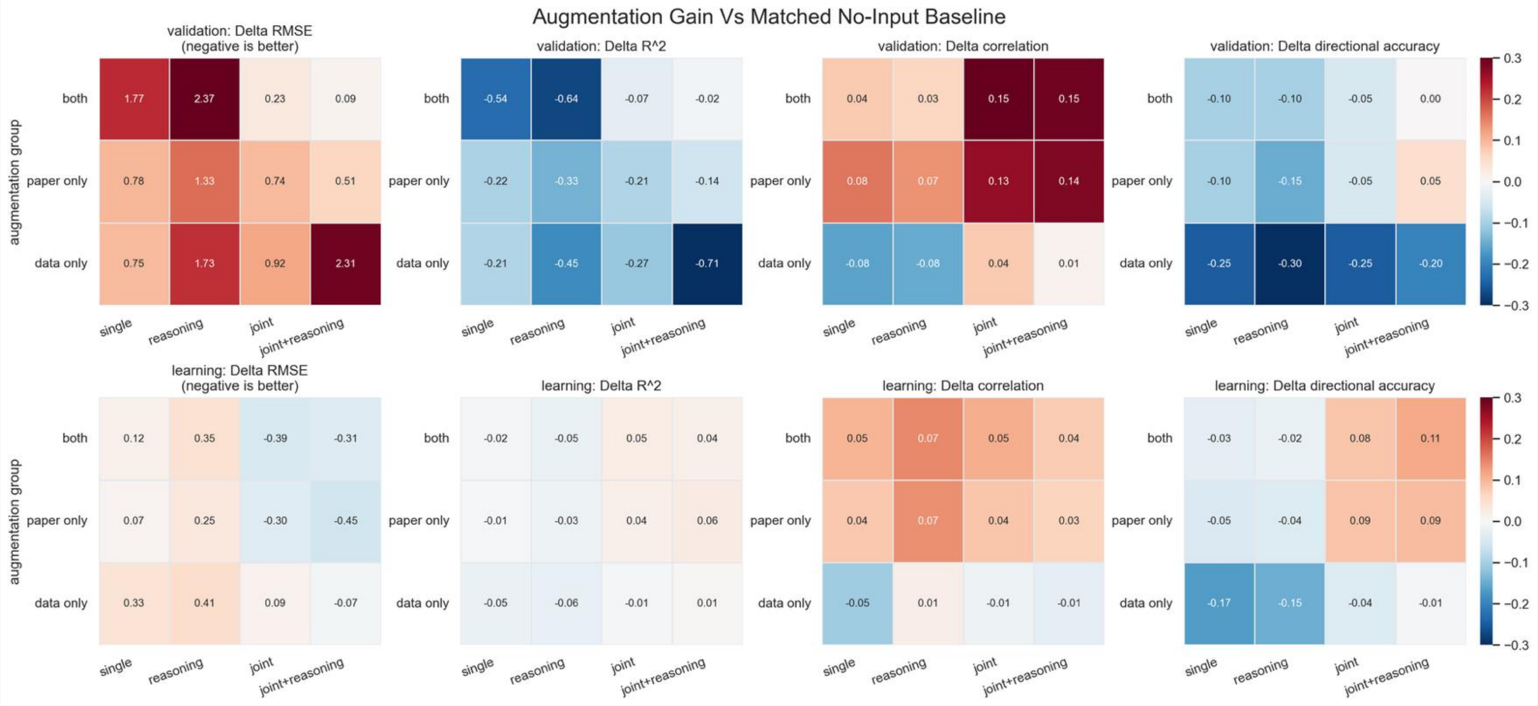
Positive-case prompt variants are the one family with positive median Δcorrelation (+0.027), but their median Δdirectional accuracy is -0.10.

Directional-accuracy gains are sparse across the whole validation search space.

On validation, any upside is much easier to find in correlation than in directional accuracy.

Where Augmentation Helps

Grouped by input family and elicitation mode across validation and learning waves.



Validation: gains are limited and mostly concentrated in correlation rather than RMSE/R^2.

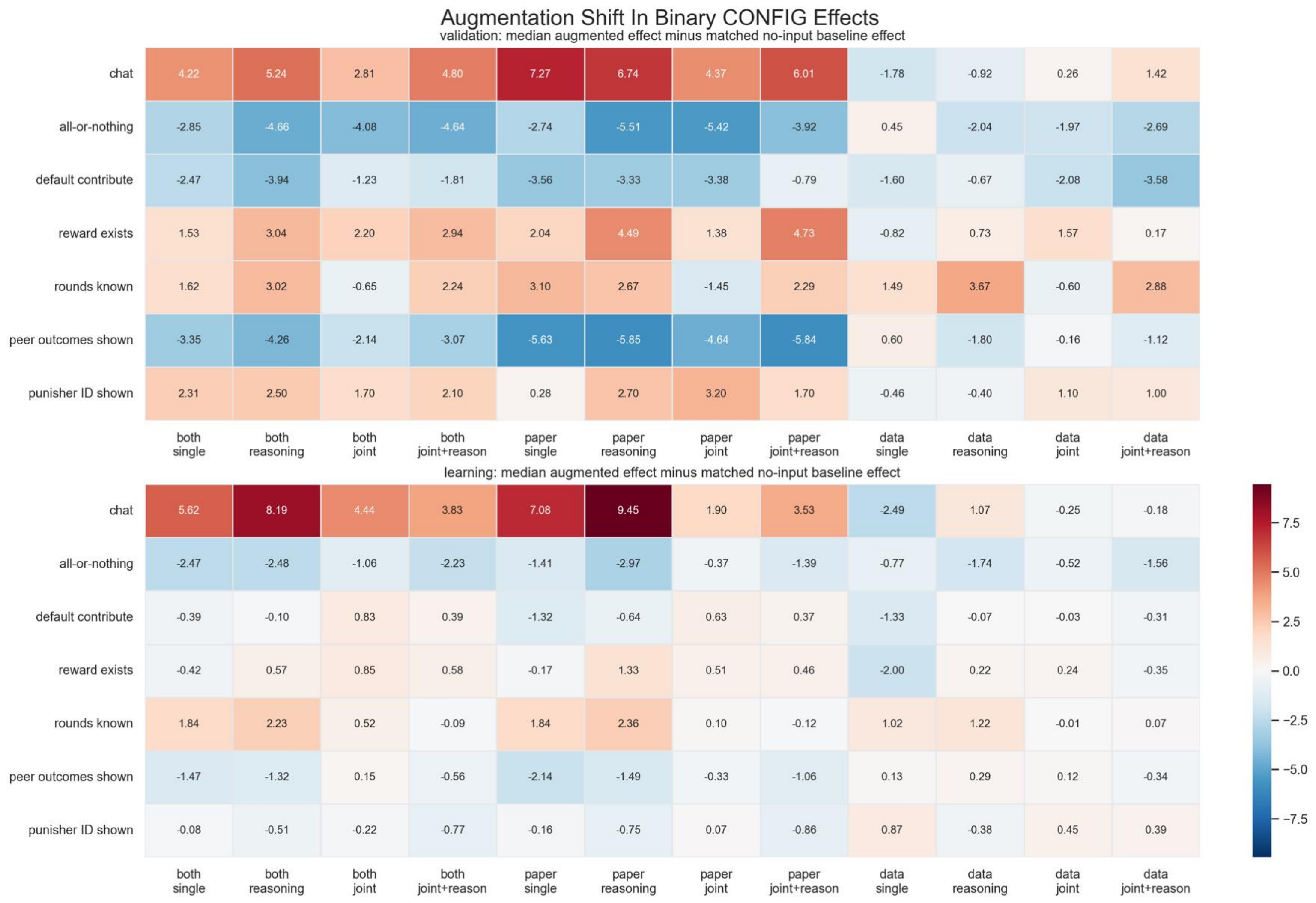
Learning: both + joint is the strongest GPT-4.1 region; paper_only can also help.

Data_only is consistently weak across waves and metrics.

Augmentation helps only in selective regimes, not as a general recipe.

Augmentation Changes The Model's Priors

Cell values show how augmentation changes the predicted treatment-effect contrast associated with each binary CONFIG.



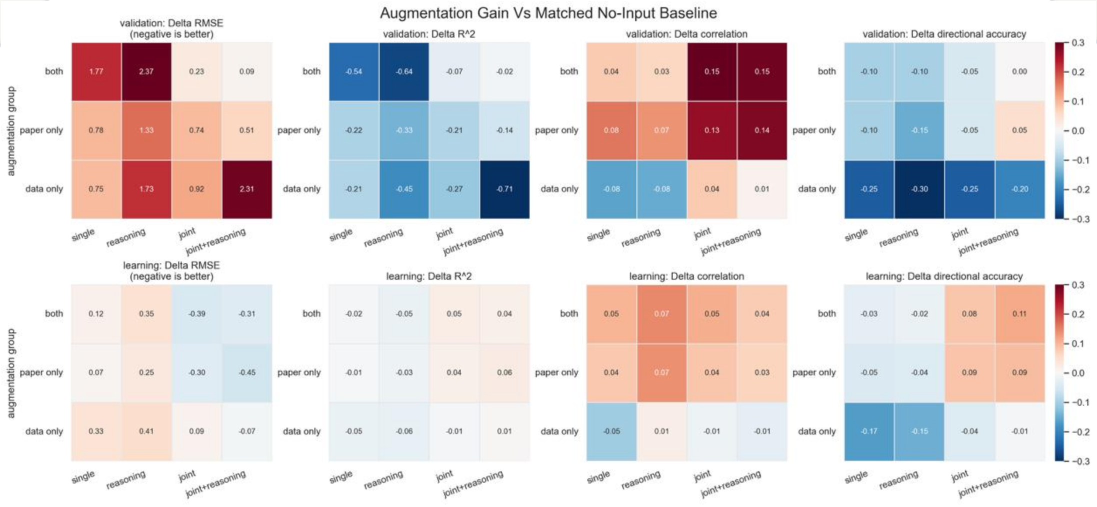
Changes are substantive, not cosmetic: augmentation shifts the model's response to chat, rewardExists, and showOtherSummaries.

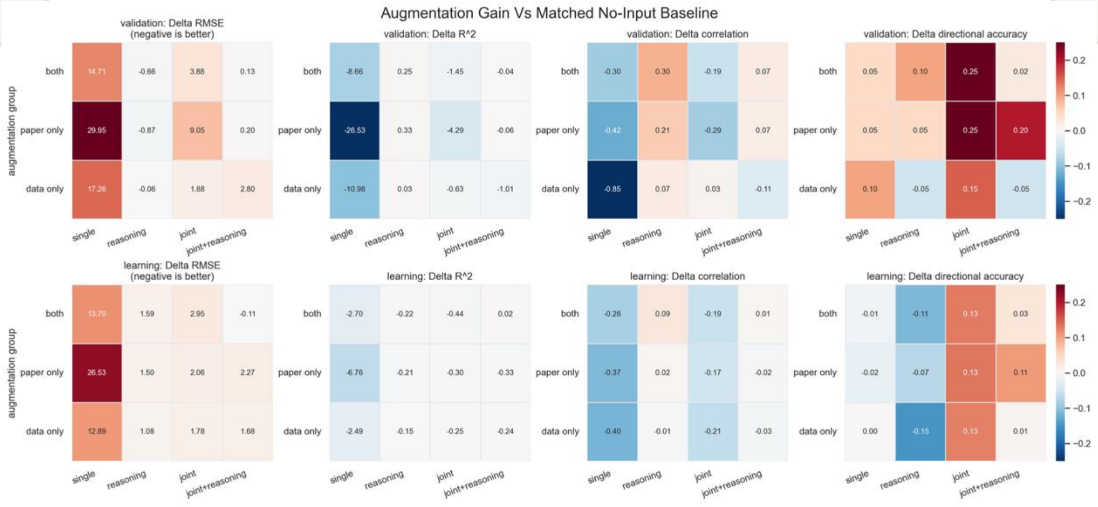
These shifts depend on both input family and elicitation mode.

Performance differences and belief changes move together; augmentation is changing what the model keys on.

Nano Is Not A Drop-In Substitute

The same analysis pipeline on GPT-4.1-nano produces a materially different pattern.





GPT-4.1 cross-wave search has 0 missing outputs; GPT-4.1-nano has 71 missing outputs, including 50 missing learning cases for both_refined_joint_reasoning.

Model coice changes the story: nano should be treated as a separate regime, not a cheaper replica.

Some coarse trends survive, especially data_only being weak.

But the winning regions change and several learning-wave gains disappear under nano.

Takeaways

On validation, the canonical GPT-4.1 baseline remains hard to beat even after a large augmentation search over report, RAG, abstract, and prompt variants.

Within unaugmented prompting, GPT-4.1 baseline_reasoning is the strongest simple setup; elastic net still outperforms the LLM baselines on R^2.

Augmentation is only useful in selective regimes, mainly with both/paper_only inputs; data_only is consistently weak.

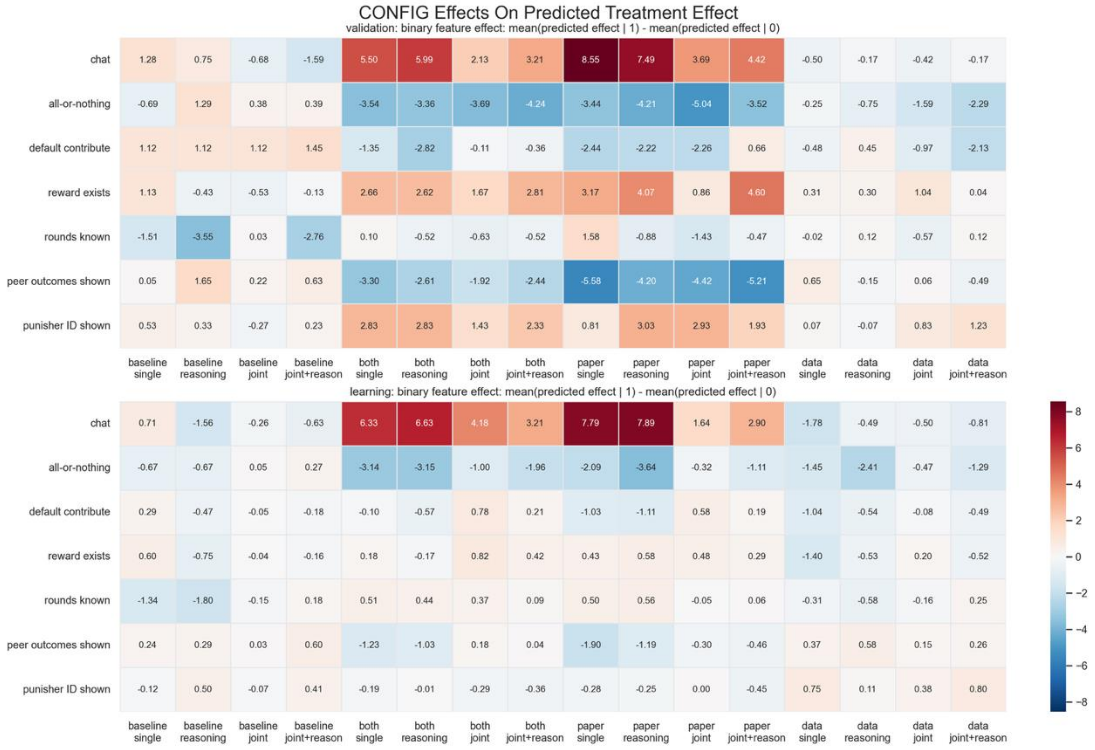
Prompting and augmentation change the model's implied causal beliefs about CONFIGs, especially chat, rewardExists, and showOtherSummaries.

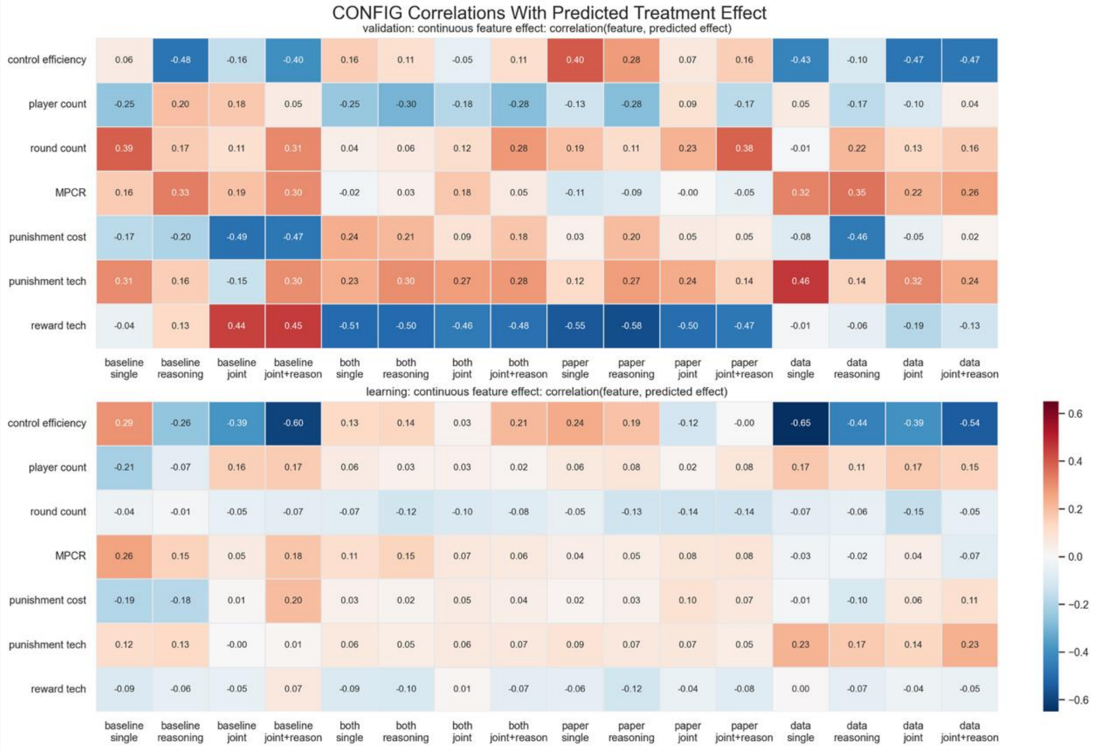
GPT-4.1-nano should be treated as a separate regime: weaker validation baselines, 71 missing outputs, and materially different augmentation patterns.

Family-level patterns are more trustworthy than exact winner rankings.

Appendix: Raw CONFIG Tendency Maps

These show the model's implied treatment-effect tendencies before differencing against the matched no-input baseline.





Binary features summarize sign and magnitude shifts for yes/no CONFIGs.

Continuous features summarize correlations between predicted treatment effect and CONFIG intensity.

The augmentation-delta slide in the main deck is the cleaner causal comparison; these are the raw tendency maps behind it.